

Synthetic example: corridor travel time

review-required. 2024 dollars. BCR 1.594; NPV \$1,188,105.

Annual economic cash flows

Year	Benefits	Costs	Net	Discounted net
2026	\$0	\$2,000,000	-\$2,000,000	-\$2,000,000
2027	\$0	\$0	\$0	\$0
2028	\$327,000	\$5,000	\$322,000	\$281,247
2029	\$327,000	\$5,000	\$322,000	\$262,848
2030	\$327,000	\$5,000	\$322,000	\$245,652
2031	\$327,000	\$5,000	\$322,000	\$229,582
2032	\$327,000	\$5,000	\$322,000	\$214,562
2033	\$327,000	\$5,000	\$322,000	\$200,525
2034	\$327,000	\$5,000	\$322,000	\$187,407
2035	\$327,000	\$5,000	\$322,000	\$175,147
2036	\$327,000	\$5,000	\$322,000	\$163,688
2037	\$327,000	\$5,000	\$322,000	\$152,980
2038	\$327,000	\$5,000	\$322,000	\$142,972
2039	\$327,000	\$5,000	\$322,000	\$133,619
2040	\$327,000	\$5,000	\$322,000	\$124,877
2041	\$327,000	\$5,000	\$322,000	\$116,708
2042	\$327,000	\$5,000	\$322,000	\$109,073
2043	\$327,000	\$5,000	\$322,000	\$101,937

Year	Benefits	Costs	Net	Discounted net
2044	\$327,000	\$5,000	\$322,000	\$95,268
2045	\$327,000	\$5,000	\$322,000	\$89,036
2046	\$327,000	\$5,000	\$322,000	\$83,211
2047	\$327,000	\$5,000	\$322,000	\$77,767

Method, sources and review

Calculation status: review-required. Responsible analyst review and program acceptance are not established by this calculation.

Project: cd275071-340a-4928-a6fa-f7f25bf0f8bc

Program: USDOT discretionary grants, 2026 method

Edition: USDOT December 2025 / 2026 update

Method source: <https://www.transportation.gov/sites/dot.gov/files/2025-12/Benefit%20Cost%20Analysis%20Guidance%202026%20Update%20%28Final%29.pdf>

Controlling program: Verify the current notice

Standalone BCA narrative and unlocked calculation workbook.

Full project capital costs, regardless of funding source; incremental No Build comparison.

O&M changes and residual value in the BCR numerator; upfront capital in the denominator.

Show physical effects that are not monetized separately.

Applicability: Training example only. No application eligibility is asserted.

Scope

Synthetic corridor

No Build: Synthetic No Build case with 100,000 annual person-hours.

Build: Synthetic Build case with reduced person-hours. No real project or traffic observation.

Method

Annual Build/No Build differences in constant 2024 dollars. End-of-year discounting to 2026.

Dollar year 2024; discount epoch 2026; real rate 7%; analysis 2026 to 2047; opens 2028.

Reduction benefits = (No Build minus Build) × annualization × unit value × dollar conversion. Increase benefits, including residual value, use Build minus No Build. Costs reverse the quantity difference. Annual overrides replace the base quantities and unit value for that year. Growth applies only in years without an override. All flows occur at year end; the discount epoch has exponent zero.

BCR uses capital in the denominator. Net operating/rehabilitation costs reduce the numerator. Residual is entered as a final-year benefit.

Results

BCR: 1.594

Net present value: \$1,188,105

Present-value numerator: \$3,188,105

Present-value denominator: \$2,000,000

Project elements

Allocate shared costs and effects once. Element ratios are not additive.

Whole project: BCR 1.594; NPV \$1,188,105; 3 open checks.

Sensitivity

Illustrative stress ranges. Replace with project-specific evidence; these are not confidence bounds.

These scenarios are assumptions, not confidence intervals.

Base case: BCR 1.594; NPV \$1,188,105.

Lower benefits: BCR 1.270; NPV \$540,583.

Higher benefits: BCR 1.918; NPV \$1,835,627.

Lower costs: BCR 1.774; NPV \$1,393,056.

Higher costs: BCR 1.220; NPV \$573,254.

Lower benefits and higher costs: BCR 0.971; NPV -\$74,268.

3% discount rate: BCR 2.326; NPV \$2,651,016.

Review worklist

- review: Capital cost: Synthetic training assumptions is assumption or lacks a reference, locator or method.
- review: Travel time: Synthetic training assumptions is assumption or lacks a reference, locator or method.
- review: Incremental maintenance: Synthetic training assumptions is assumption or lacks a reference, locator or method.

Benefits not monetized

None recorded. This does not establish that none exist.

Exclusions

Synthetic example excludes safety, emissions, induced demand and residual value.

Analyst review

Not recorded. No approval is inferred.

Sources

- example: Synthetic training assumptions. OpenPlan synthetic example, Training example, no observed data. Status: assumption. Method: Illustrative inputs for inspecting arithmetic. Limitation: Not a real project. Do not submit.. Owner: unassigned; due: unset.

Reuse

OpenPlan supporting analysis. This is not a native Cal-B/C workbook and does not execute its macros. Review required program forms separately. The XLSX has editable annual inputs and formulas; source/review checks are a snapshot and do not refresh when Excel inputs change. Return changes to OpenPlan for a new review and retained version.